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Ze-Qi decoction inhibits non-small cell lung cancer growth and metastasis by modulating the PI3K/Akt/p53 signaling pathway

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ABSTRACT

Background: The Ze-Qi decoction (ZQD) is a traditional Chinese herbal formula commonly applied to treat lung cancer in China. This study aimed to assess the effective ingredients and molecular mechanisms of ZQD in treating non-small cell lung cancer (NSCLC) based on network pharmacology combined with experimental validation.

Methods: Network pharmacology, bioinformatics, and molecular docking analyses were conducted to explore the mechanism of ZQD for treating NSCLC, which was further confirmed by animal experiments. **Results:** In total, 117 bioactive ingredients and 499 target proteins of ZQD were identified. Network pharmacology revealed 7 core active ingredients and 74 core target proteins. Kyoto Encyclopedia of Genes and Genomes enrichment analyses indicated that the PI3K/Akt and p53 signaling pathways may be crucial in NSCLC treatment. Molecular docking analysis revealed that the seven crucial bioactive ingredients complexed with PI3K, Akt, and p53. The animal experiment results validated that ZQD treatment promoted cell apoptosis and cell cycle arrest, thereby inhibiting NSCLC growth and metastasis. Furthermore, ZQD treatment caused a significant increase in p53 and Bax, while leading to a distinct reduction in p-PI3K (Tyr317), p-Akt (Ser473), VEGFA, CD31, MMP2, MMP9, Bcl2, and CDK2.

Conclusions: ZQD inhibited the growth and metastasis of NSCLC subcutaneous tumors in C57BL/6J mice via the PI3K/Akt/p53 signaling pathway.

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1. Introduction

Lung cancer, the primary cause of cancer-related deaths worldwide, is characterized by a high incidence and mortality rate.

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According to a report, approximately 2.2 million new cases and 1.8 million lung cancer deaths occurred in 2020.¹ Non-small cell lung cancer (NSCLC) accounts for 85% of all lung cancer cases,² and a large proportion of patients with NSCLC are diagnosed at late stages, presenting metastases, since early symptoms are not evident and NSCLC has a high rate of metastasis. Thus, the treatment of NSCLC still poses a significant clinical challenge. In recent decades, the therapeutic alternatives for NSCLC, such as neo-adjuvant chemotherapy, targeted therapy, and immunotherapy, have been continuously refined,^{3,4} thereby improving clinical outcomes and survival to some extent. However, the 5-year survival rate of patients with NSCLC remains below 20%,⁵ and the treatment outcomes are unsatisfactory. Therefore, novel therapeutic options are urgently needed.

List of abbreviations

NSCLC	Non-small cell lung cancer
TUNEL	transferase-mediated dUTP-biotin nick-end labeling
WB	Western blotting
ECM	extracellular matrix
MMP	matrix metalloproteinase
Bax	Bcl2 associated X-protein
Bcl-2	B-cell lymphoma 2
VEGFA	vascular endothelial growth factor A

Traditional Chinese medicine (TCM) provides a promising method for treating complex diseases like cancer due to its multi-ingredient, multi-target, and multi-functional pharmacological features.⁶ As an adjuvant therapy, TCM has been demonstrated to enhance the efficacy of radiotherapy and chemotherapy, reduce adverse reactions, prolong patient survival, and improve their quality of life.^{7,8} The Ze-Qi decoction (ZQD) is a well-known Chinese formula from *Treatise on Febrile and Miscellaneous Diseases* (Shang Han Za Bing Lun). It consists of nine herbs, namely *Pinellia ternata* (Ban-Xia, BX), *Salvia chinensis* (Zi-Shen, ZS), *Euphorbia helioscopia* (Ze-Qi, ZQ), *Scutellaria baicalensis* (Huang-Qin, HQ), *Rhizoma Cynanchi Stauntonii* (Bai-Qian, BQ), *Rhizoma Zingiberis Recens* (Sheng-Jiang, SJ), *Ramulus cinnamomi* (Gui-Zhi, GZ), ginseng (Ren-Shen, RS), and licorice root (Gan-Cao, GC). In China, ZQD has been reported to treat respiratory diseases, including acute and chronic bronchitis, bronchial asthma, bronchiectasis, as well as lung cancer, owing to its ability to protect lung function and reduce discomfort. A previous study demonstrated that ZQD could inhibit A549 and H460 cell proliferation by inducing apoptosis and cell cycle arrest.⁹ Xu, Zhu¹⁰ revealed that ZQD could improve the immunosuppressive microenvironment by regulating the STAT3/S100A9/Bcl2/caspase3 signaling pathway, causing myeloid-derived suppressor cell (MDSC) apoptosis. Beyond that, numerous studies have reported that effective components of ZQD exert different anti-NSCLC activities, including enhancing killing activities and immune responses against tumor cells,¹¹ inducing tumor cell autophagy,¹² inhibiting angiogenesis, energy metabolism,¹³ and epithelial-mesenchymal transition (EMT),¹⁴ ultimately halting tumor cell invasion and metastasis.¹⁵ Due to the multi-ingredient and multi-target pharmacological properties of TCM, studies on a single component have not been sufficient to fully elucidate the potential mechanism of ZQD in NSCLC treatment. Thus, a comprehensive and systematic method is needed to further research the bioactive compounds and molecular mechanisms of ZQD against NSCLC.

Network pharmacology is a valuable method to elucidate the molecular mechanisms of pharmacological therapies through component-target-pathway interaction network analysis, which resembles the theoretical framework of the TCM with a holistic perspective, explaining the synergistic effects of multi-component and multi-target intervention and its association with diseases.¹⁶ Accumulating evidence demonstrates that TCM exerts its pharmacological effects through diverse targets and pathways, which might be predicted using network pharmacology.¹⁷

In the present study, we first investigate the mechanisms of ZQD in treating NSCLC by network pharmacology. Moreover, the effects of ZQD on apoptosis, cell cycle arrest, angiogenesis, ECM remodeling, as well as the safety of oral ZQD were assessed and validated *in vivo*. The results showed that the underlying anti-NSCLC action mechanism of ZQD may be mediated by PI3K/Akt/p53 signaling regulation to inhibit NSCLC growth and migration. This work

provided a reference for the future clinical applications of ZQD in NSCLC treatment.

2. Materials and methods

2.1. Active ingredients in ZQD

All herbal compounds of the nine herbs in ZQD were obtained from the Traditional Chinese Medicine Systems Pharmacology (TCMSP) database (<https://tcmsp-e.com/>), using the keywords “BanXia,” “ZiShen,” “Ze-Qi,” “HuangQin,” “BaiQian,” “ShengJiang,” “GuiZhi,” “RenShen,” and “GanCao.” Absorption, distribution, metabolism, and excretion parameters have been previously evaluated to assess the pharmacokinetic properties of these chemical compounds. In the present study, oral bioavailability $\geq 30\%$ and drug-likeness ≥ 0.18 were set as the screening criteria to identify bioactive compounds in ZQD.

2.2. Targets of the active ingredients in ZQD

TCMSP, Bioinformatics Analysis Tool for Molecular Mechanism of Traditional Chinese Medicine (BATMAN-TCM, <http://bionet.ncpsb.org.cn/batman-tcm/>), and Encyclopedia of Traditional Chinese Medicine (ETCM, <http://www.tcmip.cn/ETCM/index.php/Home/Index/>) were used to identify the putative target proteins of the active compounds of ZQD. BATMAN-TCM is an online bioinformatics tool for studying the molecular mechanism of TCM; the parameters for identifying the putative targets of ZQD were a score cutoff ≥ 20 and an adjusted *P*-value ≤ 0.05 . Constituents of the most commonly used herbs and formulations of TCM are covered in detail by ETCM, with the parameter set as Tanimoto score > 0.8 . Subsequently, all obtained genes were standardized with UniProt (<https://www.uniprot.org/>), with species limited to *Homo sapiens* to standardize gene names and organisms. Finally, after removing duplicates, all putative targets were input into Cytoscape software (version 3.7.2, <https://www.cytoscape.org/>) to establish and visualize the compound-targets network of ZQD. To identify the key active components of ZQD, the topological parameters degree centrality (DC), betweenness centrality (BC), and closeness centrality (CC) were used to select the active components in the compound-target network. The components with DC, BC, and CC values among the top ten were identified as core active components of ZQD.

2.3. Acquisition of NSCLC disease targets

Gene Expression Omnibus (GEO, <http://www.ncbi.nlm.nih.gov/geo/>) is an ever-increasing platform for gene expression datasets. Three datasets, namely GSE19188, GSE21933, and GSE74706, were collected from GEO. Differentially expressed genes (DEGs) between the normal and tumor tissues, which might exert crucial effects in the development of NSCLC, were detected with the criteria of adjusted *P*-value < 0.01 and $|\log_2(\text{fold change})| > 1$. DisGeNET (<https://www.disgenet.org/>) combines and standardizes data on disease-related genes and variations from various sources, including scientific publications. NSCLC-related genes were obtained using “non-small cell lung carcinoma” as the keyword in DisGeNET.

2.4. Protein-protein interaction (PPI) network construction

The PPI network was built with the BisoGenet plugin of Cytoscape, and the topological analysis of each node in the network was determined utilizing the CytoNCA plugin. To identify key targets, six topological parameters, namely DC, BC, CC, eigenvector

centrality (EC), local average connectivity-based method (LAC), and network centrality (NC), were used to reduce the number of nodes with lower parameter values, yielding core targets/nodes. First, genes with DC values greater than the median number were selected to form a subnetwork. Genes with DC values greater than twice the median value, as well as BC, CC, EC, LAC, and NC values greater than their corresponding median values, were then screened to build a central network and were identified as core target proteins.

2.5. Gene ontology (GO) and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses

To better understand the molecular mechanism of ZQD against NSCLC, DAVID 6.8 (<http://david.ncifcrf.gov>) was used to analyze GO function and KEGG pathway enrichment of the core target proteins. GO functions include molecular function (MF), biological process (BP), and cellular component, which are used to define and describe the functions of gene products. KEGG enrichment analysis allows recognition of signaling pathways enriched by the core targets in the anti-NSCLC action of ZQD. The results, screened using *P*-value <0.01 and false discovery rate <0.01, are shown as bubble plots. Additionally, a key pathway-target network was built by Cytoscape software based on the results of the KEGG analysis.

2.6. Molecular docking analysis

Based on the TCMSP database, mol2 files of the core active ingredients of ZQD were acquired and saved in pdbqt format after importing into AutoDockTools¹⁸ software (<http://mgltools.scripps.edu/>). Three-dimensional crystal structures of core targets were collected from the Protein Data Bank database (<https://www.rcsb.org>). Water molecules and ligands were eliminated using PyMOL software. AutoDockTools software was used to add polar and nonpolar hydrogen atoms and compute the Gasteiger charge. Molecular docking was applied by AutoDock Vina, and the partial results were visualized through PyMOL software.

2.7. Establishment of a tumor-bearing mouse model

Four-week-old male C57BL/6J mice were purchased from Jinan Pengyue Experimental Animal Co. Ltd. (China). All animal experiments were conducted according to the guidelines of the Animal Ethics Committee of the Affiliated Hospital of Shandong University of Traditional Chinese Medicine, and the ethical approval number is AWE-2022-051. At 6 weeks of age, Lewis lung carcinoma (LLC) cells (1×10^6 cells in 100 μ L phosphate-buffered saline) were inoculated subcutaneously into the right armpit of each mouse. Three days after inoculation, experimental mice were randomly divided into four groups and assigned to different treatments. The body weight and tumor size were measured starting on the first and fifth day of inoculation with LLC cells, respectively, and were measured every 3 d thereafter. Tumor volume was calculated using formula $1/2 (\text{length} \times \text{width}^2)$. Mice in the ZQD-L and ZQD-H groups received ZQD at low (ZQD-L, 8.91 g/kg/day) and high dosage (ZQD-H, 35.62 g/kg/day) gavaged daily, respectively, and those in other groups were administered the same volumes of purified water by gavage. Mice in the cisplatin group received intraperitoneal injection of 3 mg/kg cisplatin (a positive control drug for antitumor action) every 3 d from the third day after inoculation; those in the other groups received the same volumes of saline via intraperitoneal injection.

2.8. In vivo safety testing

All mice were euthanized after the *in vivo* experiments. Blood samples, as well as the heart, liver, kidneys, spleen, and lungs, were collected for analysis and evaluation. The major organs were preserved in 4% formaldehyde. For histological analysis, the fixed tissues were paraffin-embedded, sectioned into 4- μ m-thick slices, and stained with hematoxylin and eosin. Automated hematology analyzers (Mindray BC-2800Vet; Shenzhen Mindray Bio-Medical Electronics, Shenzhen, China) were used to perform blood tests, including routine blood analysis and white blood cell categorization. For serum biochemical analysis, renal and liver function tests were performed. Renal function was assessed based on blood urine nitrogen (BUN), creatinine (CREA), and uric acid (UA) levels. The liver function tests included alanine aminotransferase (ALT), aspartate aminotransferase (AST), albumin (ALB), and alkaline phosphatase (ALP).

2.9. Terminal deoxynucleotidyl transferase-mediated dUTP-biotin nick-end labeling (TUNEL) assay

Apoptotic cells were detected by TUNEL assay in the tumor tissues, following the manufacturer's instructions (C1098; Beyotime, Shanghai, China). Briefly, dewaxed sections were incubated with proteinase K at 37 °C for 0.5 h, followed by incubation with enhanced endogenous peroxidase blocking buffer (P0100B; Beyotime, Shanghai, China) at room temperature for 20 min. The slices were then incubated with terminal deoxynucleotidyl transferase and biotin-dUTP in a dark environment at 37 °C for 1 h. The termination solution was added at room temperature for 10 min and incubated with streptavidin-horseradish peroxidase (HRP) at room temperature for 0.5 h. Next, the slides were stained with diaminobenzidine, counterstained with hematoxylin, dehydrated, mounted, and observed under an optical microscope. Apoptotic cells were stained brown.

2.10. Western blotting (WB)

Proteins from the frozen tumor tissues were lysed in RIPA buffer containing 100 mM phenylmethanesulfonylfluoride fluoride and a phosphatase inhibitor cocktail. A bicinchoninic acid (BCA) protein assay kit (EC001; Sparkjade, Jinan, China) was used to quantify protein content. Protein samples were resolved by 10% SDS-polyacrylamide gel electrophoresis and transferred onto 0.2 μ m polyvinylidene difluoride membranes (Millipore, MA, USA). The membranes were blocked with 5% non-fat dry milk (9999S; Cell Signaling Technology, Danvers, MA, USA) for 2 h, then incubated with primary antibodies at 4 °C overnight. The antibodies used were rabbit anti-matrix metalloproteinase 2 (anti-MMP2; 1:1000, 10373-2-AP; Proteintech, Wuhan, China), rabbit anti-MMP9 (1:1000, 10375-2-AP; Proteintech, Wuhan, China), rabbit anti-CD31 (1:1000, ER31219; Huabio, Hangzhou, China), rabbit anti-vascular endothelial growth factor A (anti-VEGFA; 1:1000, ET1604-28; Huabio, Hangzhou, China), mouse anti-P53 (1:1000, 60283-2-Ig; Proteintech, Wuhan, China), rabbit anti-cyclin-dependent kinase 2 (anti-CDK2; 1:1000, bs-10726R; Bioss, Beijing, China), mouse anti-phosphoinositide 3-kinase (anti-PI3K; 1:3000, 67071-1-Ig; Proteintech, Wuhan, China), rabbit anti-P-PI3K (1:1000, bs-5570R; Bioss, Beijing, China), mouse anti-protein kinase B (anti-AKT; 1:20000, 60203-2-Ig; Proteintech, Wuhan, China), mouse anti-P-AKT (1:5000, 66444-1-Ig; Proteintech, Wuhan, China), rabbit anti-Bcl2 associated X-protein (anti-Bax; 1:5000, ab32503; Abcam, Cambridge, UK), rabbit anti-B-cell lymphoma 2 (anti-Bcl2; 1:2000, ab32503; Abcam, Cambridge, UK), and rabbit anti-HRP-conjugated β -actin (1:20,000, HRP-60008;

Proteintech, Wuhan, China). Next, secondary anti-mouse HRP-conjugated antibodies (1:50,000, GAM0072; Multi Sciences, Hangzhou, China) or secondary anti-rabbit HRP-conjugated antibodies (1:50,000, GAR0072; Multi Sciences, Hangzhou, China) were incubated at room temperature for 1.5 h; proteins were detected with ECL chromogenic substrate (MA0186-1; Meilun Biotech, China) and visualized with a ChemiDoc MP Imaging System (Bio-Rad, Hercules, Hercules, CA, USA).

2.11. Statistical analysis

Raw data are expressed as mean $\pm$ standard error of the mean. One-way and two-way analyses of variance were conducted using GraphPad Prism (version 8.0). Significance was defined as follows: * $P < 0.05$ and # $P < 0.05$ statistically significant; ** $P < 0.01$ and ## $P < 0.01$ highly significant; *** $P < 0.001$ and ### $P < 0.001$ very highly significant.

3. Results

3.1. Active ingredients and putative drug targets of ZQD

We identified 1,120 compounds in ZQD. There were 116 compounds in BX, 79 in ZQ, 28 in BQ, 143 in HQ, 220 in GZ, 16 in ZS, 280 in GC, 190 in RS, and 265 in SJ. After removing redundant compounds, 177 active ingredients were identified based on TCMSP. Information on the active chemical components of the nine herbs in ZQD was shown in [Supplementary Table 1](#). In addition, 499 corresponding targets of ZQD were collected based on TCMSP, BATMAN-TCM, and ETCM, and the herb-compound-targets network was constructed to present the potential interactions among the ingredients and corresponding targets in ZQD ([Fig. 1A](#)). [Fig. 1B](#) showed the core active compounds of ZQD, including stigmasterol, wogonin, quercetin, baicalin, baicalein, and kaempferol, which were considered the essential active compounds responsible for the anti-NSCLC action of ZQD.

3.2. NSCLC-related targets

DEGs of the three gene expression datasets (GSE19188, GSE21933, and GSE74709) were shown on volcano maps ([Fig. 1C](#)), and the top 10 up- and downregulated genes were presented as heatmaps ([Fig. 1D](#)). There were 34 common target proteins between bioactive ingredients of ZQD and NSCLC-related targets (including DEGs from the three datasets and NSCLC-related genes in DisGeNET) determined using a Venn diagram ([Fig. 1E](#)), which may be the essential target proteins in ZQD for NSCLC treatment.

3.3. PPI networks and functional enrichment analysis

After two-step screening, 74 core targets were obtained according to the six topological parameters of each node in the PPI network ([Fig. 2A](#)). Then, GO and KEGG analyses were performed based on these targets ([Fig. 2B–C](#)). Notably, BPs were mainly enriched in apoptosis, G1/S and G2/M transition of the cell cycle, transcription, and protein ubiquitination; the most enriched cellular components were nucleoplasm, nucleus, cytosol, and cytoplasmic protein complex; the most significantly enriched MF were protein binding, transcription factor binding, ubiquitin protein ligase binding, and enzyme binding. The signaling pathways participated in the cell cycle, carcinogenesis, PI3K-Akt signaling, MAPK signaling, FoxO signaling, p53 signaling, and adherens junction. All these results indicated that ZQD might inhibit the growth of NSCLC via multiple signaling pathways, which may be related to cell cycle arrest and apoptosis.

3.4. Key pathway-target network analysis

To better examine the mechanism underpinning of ZQD anti-NSCLC effects, a pathway-target network based on the top 20 signaling pathways of KEGG and their corresponding genes was constructed ([Fig. 2D](#)). The network included 71 nodes (51 genes and 20 pathways) and 219 edges. Among the 20 pathways, “cell cycle” and “PI3K/Akt signaling” accounted for the highest proportion, except for the “cancer pathways”. In addition, AKT1, TP53, CDKN1A, MYC, MDM2, EGFR, and MAPK1 were also highly correlated targets, suggesting they may mediate the essential therapeutic effects of ZQD on NSCLC. Notably, these key target proteins of ZQD simultaneously participate in anti-NSCLC by several pathways. For instance, AKT1 is involved in cancer pathways and PI3K-Akt signaling pathway; p53, CDKN1A, MDM2, and CDK2 all belong to pathways in cancer, PI3K-Akt signaling pathway, cell cycle, and p53 signaling pathway. These results implied that ZQD played anti-NSCLC effects via several target proteins and pathways.

3.5. Molecular docking analysis

The key active ingredients stigmasterol, wogonin, quercetin, baicalin, baicalein, and kaempferol were docked with the key targets PI3K and Akt. The results were shown in [Fig. 2E](#). All molecular docking models were built and validated as satisfactory, except for Akt-quercetin with root mean square deviation (RMSD) values greater than 2.0 Å ([Fig. 2E](#)). The partial docking conformations are presented in [Fig. 2F](#). The binding energies between the active components of ZQD and target proteins were all less than -7 kcal/mol, suggesting that the core active components of ZQD had a strong binding ability to their respective target proteins. Based on the results of network analysis and molecular docking, the core active ingredients of ZQD could tightly bind to target proteins (PI3K and Akt), preliminarily confirming the accuracy of network pharmacology and indicating that ZQD might exert anti-NSCLC effects by the PI3K/Akt signaling pathway.

3.6. ZQD suppresses the growth of NSCLC subcutaneous tumors in mice

LLC subcutaneous tumor models were established to further investigate the therapeutic efficacy of ZQD. As shown in [Fig. 3A](#), the body weight increasing trend in the ZQD group remained similar to that in controls. Moreover, the mice in the ZQD groups were heavier than in the control group, whereas the body weights of mice in the cisplatin group were significantly lighter relative to controls ($P < 0.01$, [Fig. 3A](#)). [Fig. 3B](#) showed that tumor growth was significantly suppressed by oral ZQD treatment and intraperitoneal cisplatin administration. Compared with the control group, the tumor volume was remarkably reduced in the ZQD-H group after 8 d of therapy ($p < 0.05$), which continued until the experimental end-point. Notably, the end-point tumor size in the ZQD-H group was more than 2-fold less than that of controls ($p < 0.01$, [Fig. 3B](#)). Interestingly, the subcutaneous tumors developed slower in the ZQD-H group than in the cisplatin group ([Fig. 3B](#)). The average volume and weight of tumors in the treatment groups were all lower than those in controls, and the tumors in the ZQD-H group were smaller than in the cisplatin- and ZQD-L groups, as confirmed by the results on day 21 ([Fig. 3B–D](#)).

Histologically, compared to the control group, the tumor tissues in ZQD- and cisplatin-groups showed decreased tumor cell density. Besides, the abnormal morphology of cells, nuclei, and nuclear membranes were ameliorated to varying degrees ([Fig. 3E](#)). Moreover, notable necrosis was noted in ZQD- and cisplatin-groups compared with the control group; the necrotic areas of

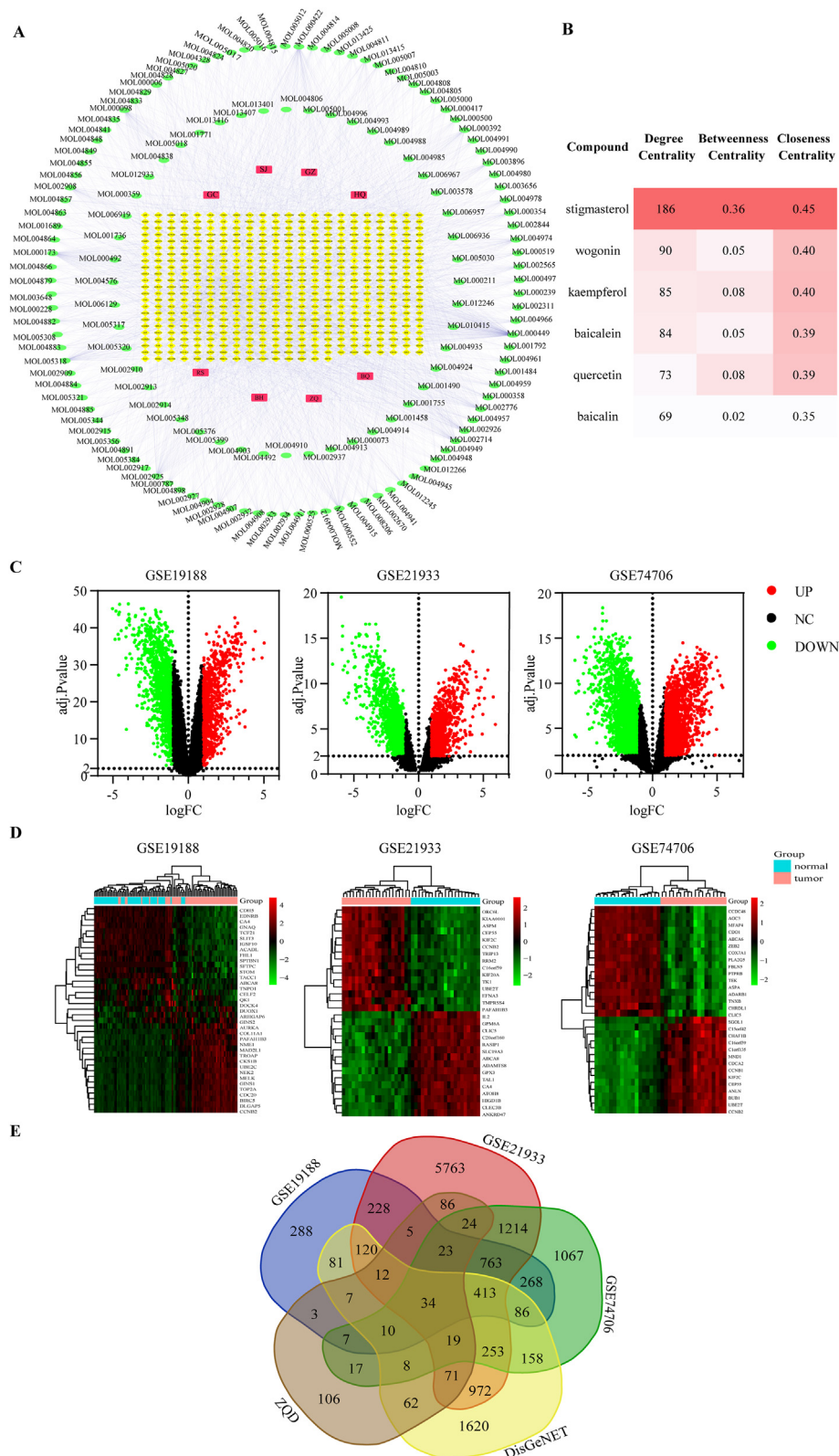


Fig. 1. Building the herb-active compound-target network of ZQD by network pharmacology and identification of the DEGs of NSCLC-related targets by bioinformatics. **(A)** A systematic network was built by connecting the bioactive components and predicted targets of the nine herbs that constitute ZQD. **(B)** Topological parameters analysis of the core ingredients in ZQD. **(C)** Volcano maps showed the DEGs of GSE19188, GSE21933, and GSE74706, which were downloaded from the GEO database. **(D)** Heatmaps showed the top 10 up- and down-regulated genes in the three GEO datasets. **(E)** Venn diagram of 34 common targets between ZQD and NSCLC-related DEGs, which were obtained from the GEO and DisGeNET databases.

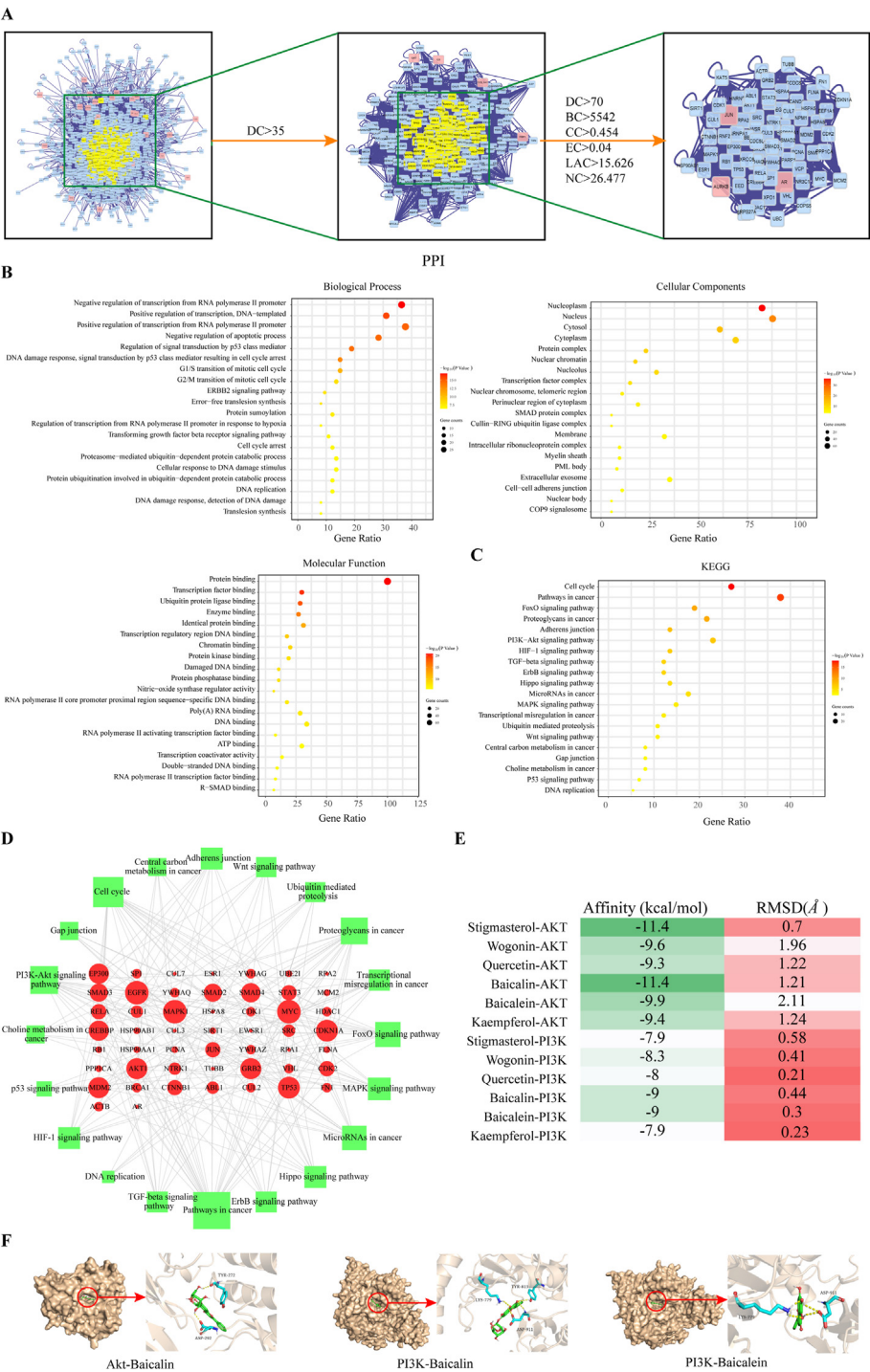


Fig. 2. Network and functional enrichment analysis of ZQD against NSCLC, as well as the preliminary validation of molecular docking. (A) The core anti-NSCLC targets of ZQD were acquired according to topological analysis parameters, including DC, BC, CC, EC, LAC, and NC. (B, C) GO and KEGG analyses were conducted to investigate the potential mechanism underlying the anti-NSCLC effect of ZQD. (D) The key pathway-target network was established according to the top 20 pathways and corresponding enriched targets in KEGG analysis. (E) The key active components of ZQD that specifically bind to PI3K and Akt, as well as RMSD method verification. The darker the green, the stronger the binding ability, and the darker the red, the more reliable the docking model. (F) Visualization of partial docking results. Abbreviations: DC, degree centrality; BC, betweenness centrality; CC, closeness centrality; EC, eigenvector centrality; LAC, local average connectivity; NC, network centrality.

specimens from the ZQD-H and cisplatin groups were larger than those from the ZQD-L group (Fig. 3E). These findings indicated that oral ZQD not only inhibited NSCLC progression but also improved the overall health status of tumor-bearing mice.

3.7. ZQD induced G1/S cell cycle arrest and apoptosis in NSCLC

Given the results of the BP analysis, we further investigated whether ZQD could intervene in cell cycle progression and induce NSCLC cell apoptosis *in vivo*. The results of the TUNEL assay indicated that the percentage of apoptotic cells in ZQD-H-, ZQD-L-, and

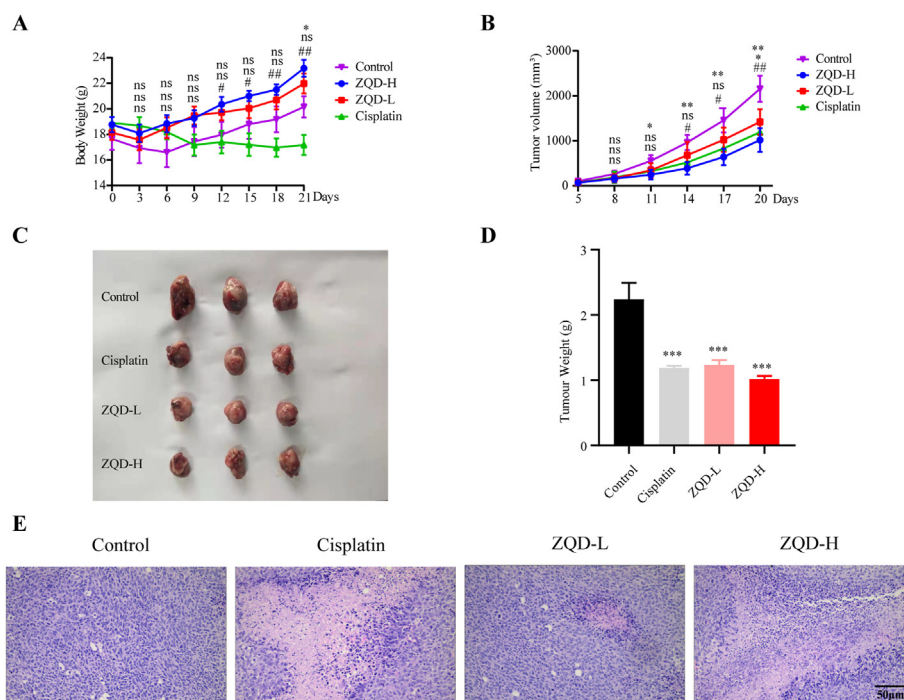


Fig. 3. ZQD inhibited NSCLC tumor growth *in vivo*. (A) After LLC cells were inoculated, the body weight of C57BL/6J mice was measured every 3 d. (B) The tumor volume was measured every 3 d starting on the fifth day after cancer cell inoculation. (C, D) Tumors were photographed and weighed after the mice were sacrificed 21 d post-inoculation. ZQD (8.91 g/kg/day, 35.62 g/kg/day) and cisplatin (3 mg/kg/3 days) treatment remarkably decreased the tumor weight compared with the control. (E) Hematoxylin and eosin staining of the tumor. *ZQD-H/L group vs. control group (ns, no significance, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$); #cisplatin group vs. control group (ns, no significance, * $p < 0.05$, ** $p < 0.01$). Values are the mean $\pm$ SEM, $n = 6$. The image is representative of three animals per experimental group (magnification $\times 200$). Abbreviations: ZQD-H, high dosage of ZQD; ZQD-L, low dose of ZQD.

cisplatin-treated tumors were considerably higher than that in controls ($p < 0.001$, $p < 0.01$, $p < 0.001$, respectively), demonstrating that ZQD effectively caused apoptosis in NSCLC (Fig. 4A). Bcl2 and Bax belong to the Bcl2 family and regulate the mitochondria-mediated apoptotic pathway. Fig. 4B showed that the expression of Bax protein expectedly increased in ZQD-H, ZQD-L, and cisplatin groups ($p < 0.001$, $p < 0.05$, and $p < 0.01$, respectively), while Bcl2 protein expression was significantly decreased in both ZQD-H and cisplatin groups (both $p < 0.01$); the level of Bcl2 in the ZQD-L group was also reduced, however, no significant difference was observed compared with the control group. CDK2, a G1/S phase-specific marker, was significantly downregulated in ZQD-treated tumors, with a remarkable difference between the ZQD-H and control groups ($p < 0.05$, Fig. 4C). Taken together, these results suggested that ZQD suppressed NSCLC growth by causing cell cycle arrest and apoptosis, further validating its antitumor effect against NSCLC.

3.8. ZQD suppresses NSCLC invasion and migration

Angiogenesis and extracellular matrix (ECM) depletion are crucial factors for tumor growth, invasion, and migration. MMPs participate in ECM degradation and promote tumor metastasis.¹⁹ MMP2 and MMP9 expression was considerably decreased in the ZQD-L, ZQD-H, and cisplatin groups compared to the control group (MMP2: all $p < 0.001$; MMP9: $p < 0.01$, $p < 0.01$, $p < 0.05$; Fig. 5A), which suggested that ZQD can suppress tumor migration and invasion. Notably, the magnitude of downregulation of MMP2 and MMP9 in the ZQD-H/L groups was larger than that in the cisplatin group, implying that the effect of ZQD in inhibiting tumor migration and invasion was superior to that of cisplatin (Fig. 5A).

VEGFA plays a key role in angiogenesis,²⁰ and CD31 is often used to assess the extent of angiogenesis in tumors and to suggest a quickly expanding tumor.²¹ In this study, the expressions of CD31

and VEGFA were markedly decreased in the ZQD-H, ZQD-L, and cisplatin groups in comparison with the control group (all $p < 0.01$) (Fig. 5B). The results indicated that ZQD inhibited angiogenesis by downregulating CD31 and VEGFA expression. Of note, ZQD-H inhibited angiogenesis to a greater extent than cisplatin. All these findings further confirmed that ZQD had direct anti-NSCLC activity *in vivo*.

3.9. ZQD inhibits NSCLC growth and metastasis through the PI3K/Akt/p53 signaling pathway

The key pathway-target network showed crucial pathways and targets of ZQD against NSCLC, among this network “pathway in cancer”, “cell cycle”, and “PI3K-Akt signaling pathway” had the highest specific gravity (Fig. 2D). Since the p53 signaling pathway is pivotal in the progression of the cell cycle and was also found in this network, the PI3K-Akt and p53 signaling pathways were chosen to further explore the mechanism of ZQD efficacy. Indeed, Fig. 6A–B showed that the expression of phospho-Akt (Ser473) and phospho-PI3K (Tyr317) was significantly reduced (p-Akt: $p < 0.05$, $p < 0.05$, $p < 0.001$; p-PI3K: $p < 0.01$, $p < 0.01$, $p < 0.001$), while that of p53 was markedly increased in cisplatin and ZQD-L/H groups (all $p < 0.001$). These findings indicated that the PI3K/Akt signaling pathway was expectedly inhibited, while the p53 signaling pathway was significantly activated after ZQD treatment. Collectively, these results implied that ZQD strongly exerted anticancer effects through the PI3K/Akt/p53 signaling axis, which was consistent with our network pharmacology results.

3.10. The safety profile of ZQD and cisplatin *in vivo*

Besides evaluating the efficacy of ZQD and cisplatin treatment in the subcutaneous model, biological safety was also performed in

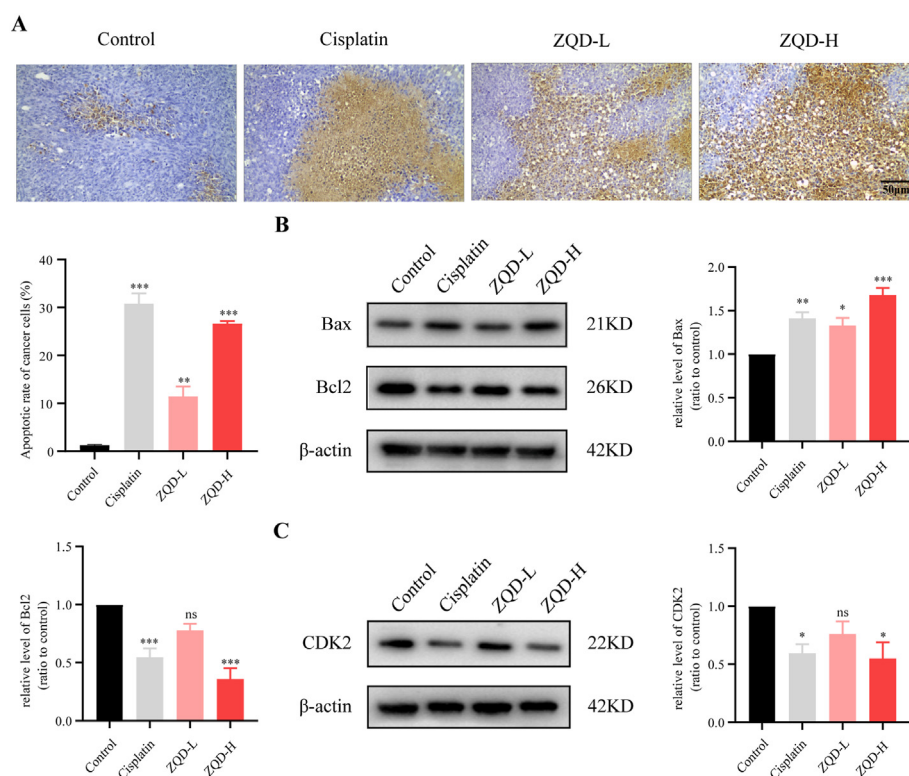


Fig. 4. ZQD caused NSCLC apoptosis and cell cycle arrest *in vivo* experiments. (A) Apoptotic rates of cancer cells based on TUNEL assay. The image is representative of three animals per experimental group (magnification $\times 200$). (B, C) Bax, Bcl2, and CDK2 relative expression in the tumor was determined by Western blot; ZQD-L/H significantly increased Bax expression while decreasing Bcl2 and CDK2 expression in NSCLC. ns, no significance, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Data represented as the mean ($\pm$ SEM) of at least three independent experiments, each performed in triplicate.

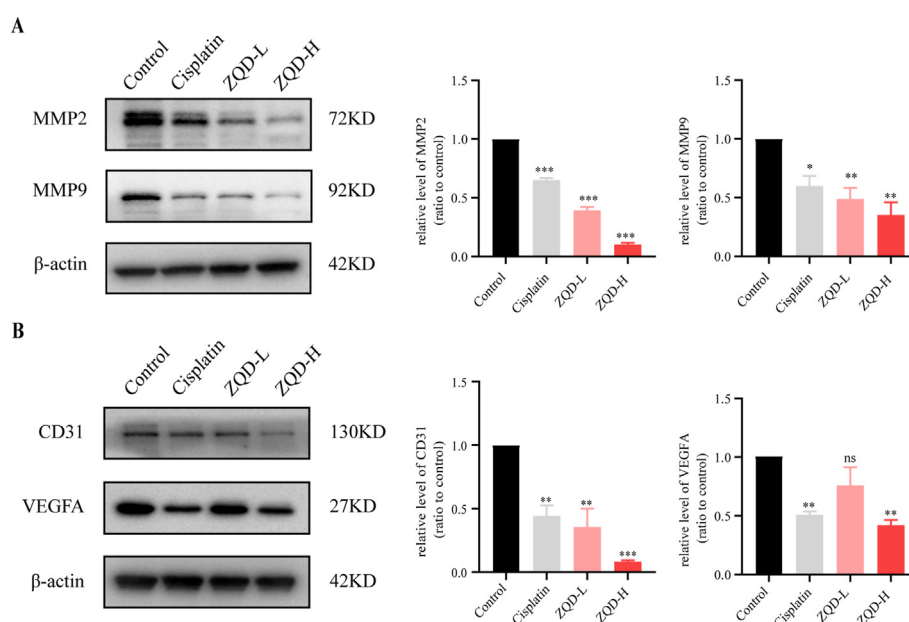


Fig. 5. ZQD inhibited NSCLC angiogenesis and metastasis *in vivo*. (A, B) Relative expression of MMP2, MMP9, CD31, and VEGFA was quantified by Western blot. MMP2, MMP9, CD31, and VEGFA were markedly downregulated after ZQD treatment. ns, no significance, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$. Data represented as the mean ($\pm$ SEM) of at least three independent experiments, each performed in triplicate.

these mice. Pathologically, compared with the control group, oral ZQD or intraperitoneal cisplatin caused no significant damage to the heart, lung, liver, spleen, or kidney of the mice (Fig. 7A). Furthermore, routine blood analyses showed no significant

difference between the treated and control groups (all $p > 0.05$, Fig. 7B). In addition, the hepatic and renal functions of mice in the treated and control groups showed no evident changes for serum ALT, AST, ALB, ALP, BUN, CREA, and UA levels (all $p > 0.05$, Fig. 7C).

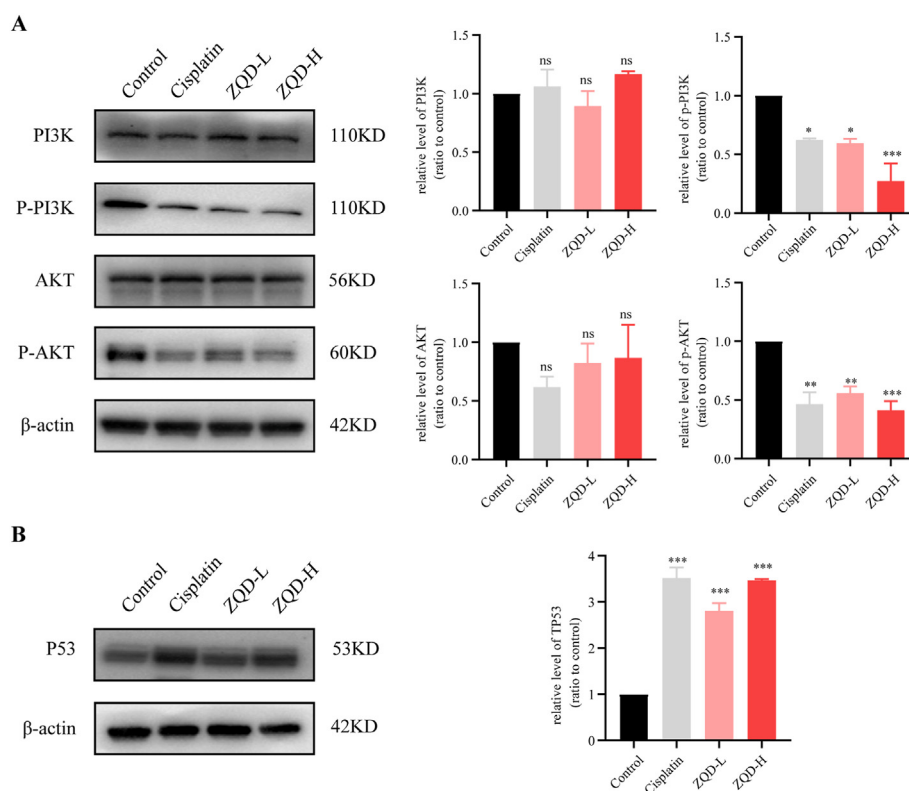


Fig. 6. ZQD regulated PI3K/Akt/p53 pathway to exert an anti-NSCLC function. **(A)** ZQD treatment led to decreased PI3K and Akt phosphorylation levels. **(B)** p53 was determined by western blot. ZQD treatment upregulated p53 expression in NSCLC. ns, no significance, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.01$. Data represented as the mean ($\pm$ SEM) of at least three independent experiments, each performed in triplicate.

These results indicated that ZQD and cisplatin caused no major organs or hematologic damage to the mice.

4. Discussion

The inconspicuous symptoms at an early stage and the propensity to metastasize are characteristics of NSCLC. Therefore, a large number of patients with NSCLC are already in the advanced or metastatic stage at the time of diagnosis,²² missing the optimal treatment time. As NSCLC growth and metastasis are complex processes involving several molecular and cellular signaling pathways, chemotherapy and targeted therapy aimed at specific BPs are prone to drug resistance, leading to unsatisfactory therapeutic efficacy^{23,24} and posing adverse side effects. Despite the emergence of immunotherapy having drastically changed the therapeutic prospects of advanced NSCLC, regrettably, only a fraction of patients benefits from it.²⁵ Notably, TCM could inhibit NSCLC progression in a multi-pathway manner, such as inducing apoptosis, inhibiting growth and metastasis, reversing multidrug resistance, and regulating immunological function, since its multi-component and multi-target properties synergistically caused a significant improvement in the prognosis of patients.²⁶

Stigmasterol, wogonin, quercetin, baicalin, baicalein, and kaempferol are core active components of ZQD that reportedly inhibit tumor growth and metastasis by regulating multiple oncogenic signaling pathways and/or biological processes in NSCLC.^{27–30} For instance, stigmasterol has been verified to trigger tumor cell apoptosis and autophagy by inhibiting the Akt/mTOR pathway.²⁷ Quercetin, a natural flavonoid, inhibits cell proliferation by reducing Bcl2, increasing Bax and caspase-3, and upregulating the SIRT1/AMPK signaling pathway in NSCLC cell lines.^{28,31} It also

suppresses the invasion and migration of A549 cells by down-regulating the TLR4/NF- κ B signaling pathway.²⁹ Through decreased expression of GST, NQO1, HO1, and Nrf2, kaempferol causes apoptosis and autophagy.^{12,32} Wogonin, baicalein, and baicalin, the primary effective ingredients of *Scutellaria baicalensis*, were identified remarkably suppressing tumor growth in LLC-bearing mice and enhancing antitumor immune responses.³⁰ In terms of wogonin, it is reported to cause apoptosis and repress cell proliferation, angiogenesis, and EMT of NSCLC by downregulating inhibitors of differentiation 1,¹⁴ decreasing HIF-1 and MCT-4 expression.^{13,33} Baicalein plays an antiproliferative role through the AMPK/mitochondrial fission pathway mediating apoptosis/autophagy.³⁴ Furthermore, baicalein has been found to prevent NCI-H446 human NSCLC cell migration by reducing p-PDK1, p-Akt, E-cadherin, and vimentin expression.³⁵ These findings showed that the core active components of ZQD played multiple roles in inhibiting NSCLC progression. The *in vivo* experiments in this study also validated that ZQD suppressed the growth of NSCLC subcutaneous tumors and that its anti-NSCLC effect outperformed that of cisplatin.

Based on network pharmacology analysis, apoptosis, transcription, and cell cycle arrest were the BPs most affected by ZQD treatment. Inducing apoptosis and cell cycle arrest are crucial mechanisms for inhibiting cancer proliferation. TUNEL assay results further confirmed the proapoptotic effect of ZQD. In addition, in this study, we found that ZQD could mediate the mitochondrial apoptosis pathway to induce NSCLC apoptosis by upregulating Bcl2 and downregulating Bax expression. Furthermore, CDK2, a key protein that affects the G1/S transition and S-phase progression in the cell cycle, and decreased CDK2 expression reportedly cause cell cycle arrest in NSCLC cell lines.³⁶ Therefore, ZQD inhibited tumor

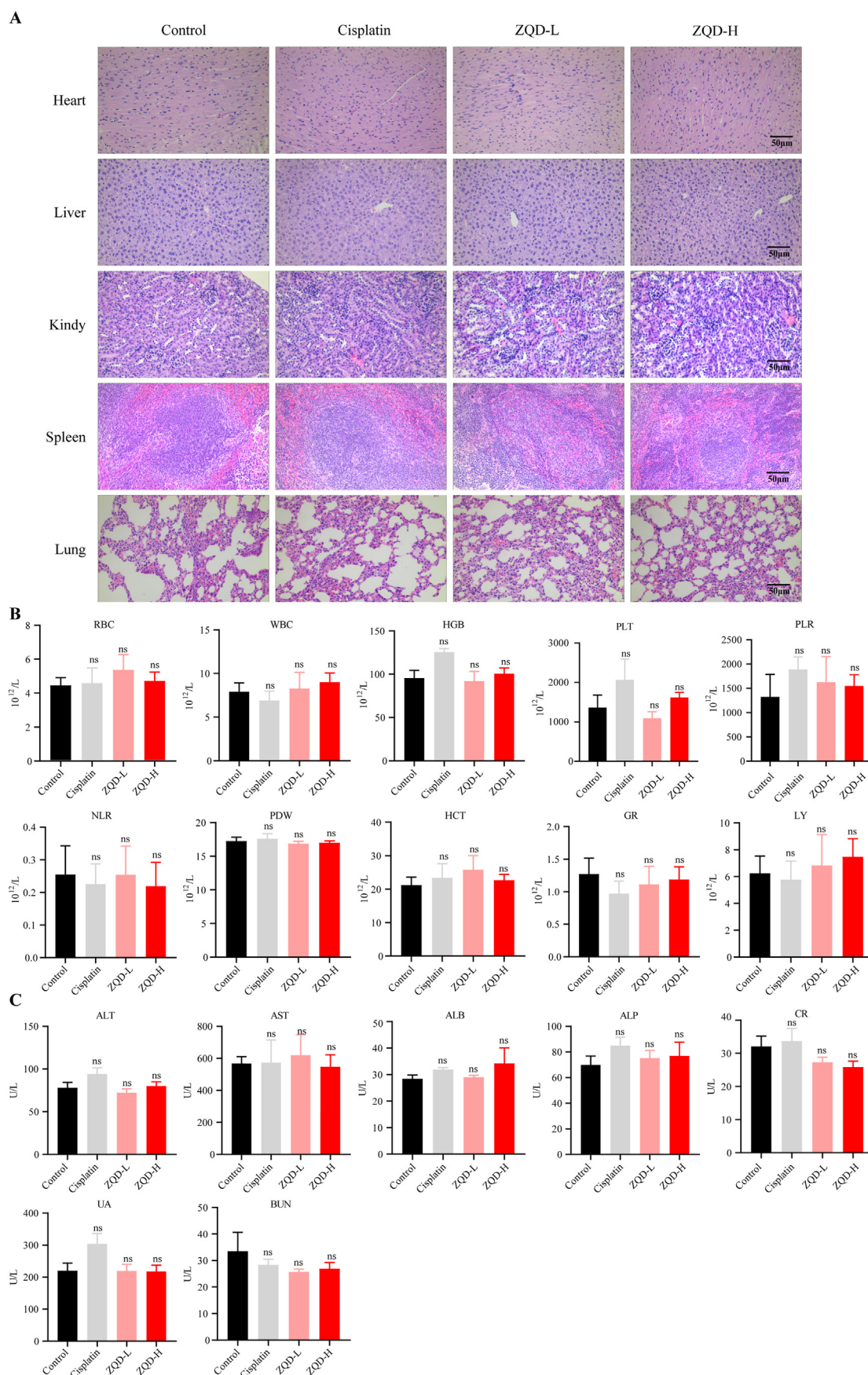


Fig. 7. ZQD caused no toxicity to tumor-bearing mice. (A) Hematoxylin and eosin staining of the heart, liver, kidney spleen, and lung. The image is representative of three animals per experimental group (magnification $\times 200$). The results of routine blood tests (ns, no significance) (B); liver and kidney function tests (ns, no significance) (C). Values are the mean $\pm$ SEM, $n = 5$.

growth and may be related to inducing cell cycle arrest, showing the therapeutic potential of ZQD as an anti-NSCLC treatment. Overall, the therapeutic role of ZQD in NSCLC may be through apoptosis induction and cell cycle arrest caused by downregulating CDK2 and Bcl2 and Bax upregulation.

Clinically, metastases remain the leading cause of NSCLC deaths among patients. Therefore, inhibiting the metastases could greatly improve the outcomes of NSCLC patients. Degrading the ECM is one of the crucial steps for tumor migration. The overexpression of MMPs, such as MMP2 and MMP9, promotes the degradation of ECM to accelerate the progression of metastatic NSCLC. In this work, the modulation of ZQD on the ECM was examined. Interestingly, ZQD reversed the elevation of MMP2 and MMP9 levels observed in NSCLC, suggesting that it inhibited NSCLC metastasis, at least partly, via an anti-degradation effect on the ECM. In addition to the degradation of the ECM, tumor metastasis is inseparable from angiogenesis, a process that includes the formation of new blood vessels from existing ones, and vasculogenic mimicry, a micro-vessel-like structure formed by cancer cells. Targeting angiogenesis is emerging as a clinically important approach to delay or suppress NSCLC progress, achieving promising results. In our experiments, VEGFA and CD31 levels were measured as markers of angiogenesis. As shown in Fig. 5, increased VEGFA and CD31 levels were observed in the control group, and the elevation of angiogenesis was attenuated by ZQD. In this study, MMP2, MMP9, CD31, and VEGFA were downregulated in NSCLC subcutaneous tumors after ZQD administration, suggesting that ZQD may inhibit NSCLC migration, invasion, and angiogenesis by suppressing MMP2, MMP9, CD31, and VEGFA expression.

According to the results of KEGG analysis, the PI3K/Akt and p53 signaling pathways play important roles in inhibiting NSCLC. Many oncogenes can activate PI3K, whose overexpression is considered a hallmark of cancer. Akt is a pivot kinase in the PI3K/Akt signaling pathway that participates in multiple functions and biological processes, including the p53 signaling pathway, cell proliferation, apoptosis, and cell cycle. p53, a nuclear phosphoprotein that is the downstream target of the PI3K/Akt signaling pathway, is involved in cell cycle arrest, DNA repair, apoptosis, angiogenesis, and metastasis and is extensively recognized as a crucial tumor inhibitor in cells.³⁷ Moreover, various studies have demonstrated that the PI3K/Akt/p53 signaling pathway is highly connected to cancer development, and blocking the PI3K signaling axis may be a promising anticancer therapeutic strategy.^{38,39} In this work, we found that the key bioactive components of ZQD not only specifically bind to PI3K and Akt but also impacted their phosphorylation level and p53 expression. Furthermore, the crucial ingredients of ZQD, such as wogonin, quercetin, baicalin, and baicalein, could decrease Akt phosphorylation, suppress the PI3K/Akt signaling pathway,^{35,40,41} and upregulate p53, exerting proapoptotic effects in NSCLC.⁴² All the research was consistent with our prediction and experiment that ZQD treatment caused the increase of p53 expression while simultaneously leading to decreased PI3K and Akt phosphorylation in NSCLC tissues. Thus, PI3K/Akt/p53 may be the primary pathway involved in the anti-NSCLC efficacy of ZQD.

Bcl2 family proteins regulate the apoptosis process through antagonistic interaction, for example, antiapoptotic protein Bcl2 inhibits the accumulation and oligomerization of the proapoptotic protein Bax, thus suppressing the permeabilization of mitochondrial outer membrane and the apoptotic process. Studies have indicated that Bax and Bcl2 are downstream effectors of p53; thus, decreased Bcl2 and increased Bax expression contribute to p53-mediated apoptosis.⁴³ In addition, p53 can induce cell cycle arrest by regulating CDK2.⁴⁴ Consequently, ZQD may exert suppressive effects on NSCLC growth by regulating the p53 signaling pathway to trigger apoptosis and cell cycle arrest. Notably, Akt and p53 exert

important effects on angiogenesis, tumor cell invasion, and migration via the regulation of VEGFA and MMPs.^{45,46} Thus, the effect of ZQD in anti-NSCLC metastasis may be caused by the decrease of MMP2, MMP9, CD31, and VEGFA expression via the Akt/p53 signaling pathway. Overall, this research showed that ZQD could significantly suppress the growth and metastasis of NSCLC subcutaneous tumors via the PI3K/Akt/p53 signaling pathway. However, ZQD could regulate also other pathways, including the MAPK signaling pathway, HIF-1 signaling pathway, Hippo signaling pathway, and TGF-beta signaling pathway which also could explain its anti-NSCLC activity according to our prediction (Fig. 2D). Multiple studies have shown that all these pathways were regulated by the core bioactive components of ZQD, such as wogonin,^{47,48} kaempferol,⁴⁹ baicalein,⁵⁰ and quercetin.^{51–53} Therefore, the comprehensive understanding mechanism of ZQD anti-NSCLC is warranted further studies to validate these predictions.

This research has certain limitations. First, since databases report incomplete information and are frequently updated, part of the bioactive components of ZQD and corresponding target proteins may not have been identified in this study. Although this preliminary study discovered some important targets and signaling pathways of ZQD in treating NSCLC, the specific upstream and downstream regulatory molecules remain unclear. Intensive investigations are required to investigate the effects of these pathways in the treatment of NSCLC. Despite these limitations, this study offers preliminary data for further studies on the clinical applications of ZQD against NSCLC.

5. Conclusion

This study employed network pharmacology, bioinformatics, and molecular docking techniques to elucidate the molecular mechanism underlying the therapeutic role of ZQD in NSCLC treatment and validated it by *in vivo* experiments. The results suggest that ZQD might inhibit angiogenesis, remodel ECM, and induce cell cycle arrest and apoptosis to suppress NSCLC growth and migration possibly via the PI3K/Akt/p53 pathways. These findings provide preliminary experimental validation of the mechanism of ZQD in treating NSCLC in the clinic.

Authors' contributions

Jingtao Zhang conducted the experiments, analyzed the data, and wrote the manuscript; Zifan Zhuang, Minghao Guo, Qingfeng Yang, and Xin Min assisted Jingtao Zhang in performing the experiments; Zifan Zhuang and Kai Wu were responsible for immunohistochemistry and TUNEL staining; Wenqiang Cui and Fei Xu designed the study, supervised the work, and revised the manuscript. All authors read and approved the final manuscript.

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Availability of data and materials

The data used for this study are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare no competing interests.

Acknowledgments

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jtcme.2023.03.008>.

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